BIFlow BO3 Data Quality Report

Full cleaning run summary generated from on-disk artifacts.

Dataset	04_dirty
Run id	up_97d9f0b8_04_dirty_csv
Generated	2026-10-26 09:00 UTC
Source	04_dirty.csv (1b4f4cccede3...)
Shape	4 × 10
LLM	LLM off
Config hash	—
Versions	Python 3.14.6, pandas 3.1.0, sklearn 1.9.1, reportlab 5.1.0

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Executive summary

PASS

Rows in	4	Rows out	4
Duplicates removed	0	Cells imputed	0
Outliers flagged	0	Review items	0
Silent deletions	0	Runtime ms	—

0 exact duplicate rows were removed and logged where applicable.
0 cells were imputed using configured deterministic or ML methods.
0 outlier flags were applied on numeric columns per YAML thresholds.
0 items remain in the review queue for human decision.

Safety & integrity checks

Invariant checks derived from parquet files and logs (never hardcoded).

Check	Status	Evidence
Row accounting 04_dirty	PASS	in=4 dropped=0 out=4
Dtype preservation 04_dirty	PASS	all original dtypes compatible
No literal 'nan' strings 04_dirty	PASS	none
Undo round-trip	NOT RUN	Full undo verification requires playground helper
Determinism	NOT RUN	No run_snapshot.json
Ingest sha256 recorded	PASS	04_dirty.csv 1b4f4cccede3...

Input & parsing

Global parse settings: {}

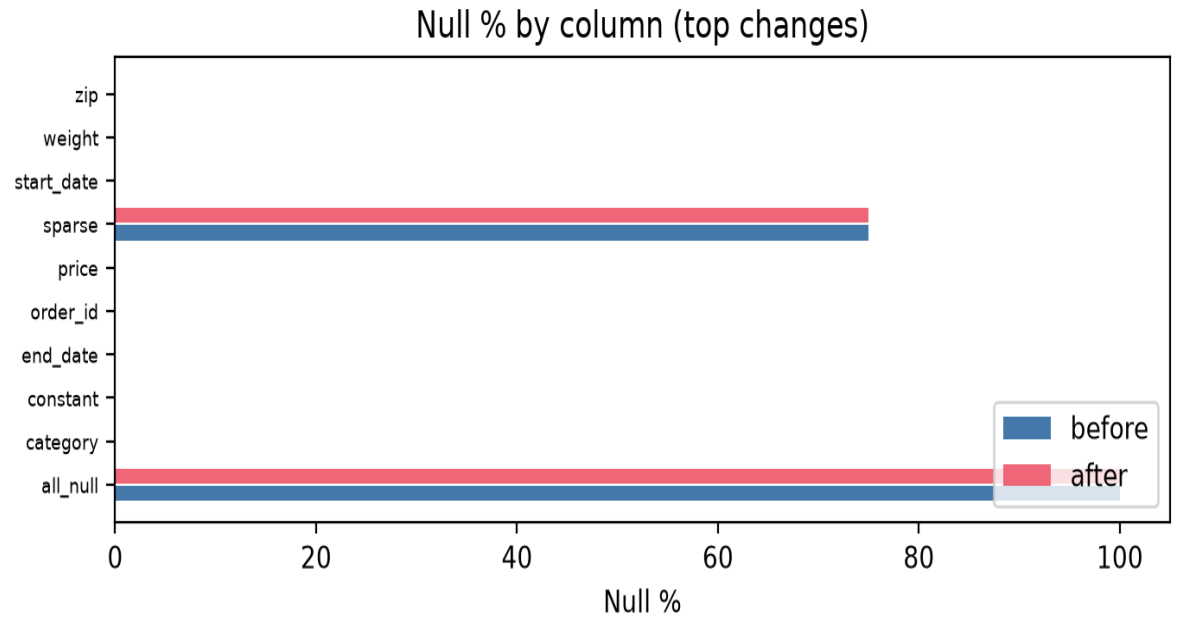
File	Encoding	Delimiter	Rows	Cols	Warnings
04_dirty.csv	auto		4	10	Column 'order_id': id-like (name matches id-like pattern (ord

Column recognition

Table	Column	Role	Type	Conf.	Action	Evidence
04_dirty	order_id	id	identifier_integer	0.92	none	id role + numeric
04_dirty	zip	categorical	postal_code	0.90	standardise_safe	postal column name
04_dirty	price	categorical	currency	0.88	review	currency-like strings in money column
04_dirty	start_date	categorical	phone	0.85	review	phone-like digits
04_dirty	end_date	categorical	phone	0.85	review	phone-like digits
04_dirty	category	categorical	categorical	0.85	review	low/medium cardinality (3 values)
04_dirty	sparse	constant	constant	1.00	standardise_safe	single distinct value
04_dirty	all_null	all_null	missing_all	1.00	standardise_safe	all values null
04_dirty	constant	constant	constant	1.00	standardise_safe	single distinct value
04_dirty	weight	categorical	categorical	0.85	review	low/medium cardinality (3 values)

Profile before vs after

Table	Rows	Dupes	Null cells (sum)
04_dirty	4	0	7



Cleaning actions

Rule	Method	Columns	Rows affected
category_map	deterministic	category	1
null_strategy	deterministic	all_null, sparse	7

Table	Column	Rule	Rows	Before	After
04_dirty	category	category_map	1	—	—
04_dirty	sparse	null_strategy	3	—	—
04_dirty	all_null	null_strategy	4	—	—

Review queue

Nothing needs review

Dropped rows & reconciliation

Table	Rows in	Dropped	Rows out	Status
04_dirty	4	0	4	PASS

Lineage

Step	Method	Rule	Rows
category_map	deterministic	category_map	1
null_strategy	deterministic	null_strategy	3
null_strategy	deterministic	null_strategy	4



LLM usage

All counts and statistics in this report are read from run artifacts; they never depend on LLM output.

LLM not used

Reproducibility & limitations

Known limitations

- Ambiguous dates go to the review queue; no silent guess.
- No semantic correction of free-text values beyond configured rules.
- Column roles block unsafe numeric imputation on ids and categoricals.
- LLM advisory (when enabled) does not change deterministic counts in this report.
- See README for supported upload formats and issue classes.

Right-to-left scripts (e.g. Arabic) are not shaped; glyphs appear in logical order only.

Appendix

Glossary

Term	Meaning
Role	Inferred column purpose (id, categorical, numeric, ...)
Flag	Boolean helper column (__imputed, __outlier, ...)
Imputation	Fill missing values with a documented statistic or model
Outlier	Values flagged by IQR or similar; not always removed
Review queue	Items needing human approval before auto-apply
Lineage	Ordered record of cleaning steps and row counts